

# Business Process Document

## AI-Based Product Similarity & Order Quantity Recommendation Tool

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| <b>Document Title</b> | AI-Based Product Similarity & Order Quantity Recommendation Tool |
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| <b>Date</b>           | 13 July 2026                                                     |
| <b>Version</b>        | 1.0 (Draft)                                                      |
| <b>Status</b>         | For Review & Feasibility Evaluation                              |

### 1. Purpose

This document outlines the business requirement for developing an AI-based decision-support tool for seasonal merchandise planning. The tool will compare upcoming season products against historical season products — using both product attributes and product images — to identify the most similar historical items, and will recommend order quantities for upcoming items based on historical order quantity, sell-through performance, and similarity analysis.

### 2. Background

Order quantity decisions for new season items are currently made using manual judgement and experience-based comparison with past products. Because attribute and performance data reside in separate files and image-based comparison is entirely manual, the process is time-consuming and inconsistent. An AI-driven solution can standardise this comparison, make historical performance directly usable at the point of decision, and reduce both over-buying and under-buying risk.

### 3. Objectives

- Compare item-wise attributes between previous seasons and the upcoming season.
- Compare product images to identify visually similar or matching styles.
- Recommend the most relevant historical items for each upcoming product, based on combined attribute and image similarity.
- Suggest a recommended order quantity for each upcoming season item, leveraging historical order quantity, sell-through performance, and similarity scores.
- Present comparison results alongside the corresponding product images in an easy-to-use interface.

## 4. Data Inputs

| # | Source File                  | Description                                                                                  |
|---|------------------------------|----------------------------------------------------------------------------------------------|
| 1 | Last Two Season Data (.xlsb) | Item-wise attribute details, order quantity, and sell-through data for the last two seasons. |
| 2 | Upcoming Season Data (.xlsb) | Item-wise attribute details for the upcoming season.                                         |
| 3 | Upcoming Season Image URLs   | Item-wise product image URLs for the upcoming season.                                        |
| 4 | Last Season Image URLs       | Item-wise product image URLs for the previous season.                                        |

## 5. Proposed Business Process

1. **Data ingestion** — Load historical and upcoming season attribute data, and fetch product images from the provided URLs.
2. **Attribute similarity** — For each upcoming item, score similarity against historical items on product attributes (e.g., category, fit, fabric, colour, pattern, price band).
3. **Image similarity** — Generate image-based similarity scores between upcoming and historical items to detect matching or near-matching styles.
4. **Match recommendation** — Combine attribute and image scores to rank and present the most relevant historical items for each upcoming product.
5. **Order quantity suggestion** — Derive a recommended order quantity for each upcoming item from the matched items' historical order quantities and sell-through performance, weighted by similarity.
6. **Review interface** — Display each upcoming item with its top historical matches, side-by-side images, similarity scores, historical performance, and the suggested order quantity, allowing the planning team to review and adjust.

## 6. Key Functional Requirements

| ID    | Requirement                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------|
| FR-01 | Item-wise attribute comparison between previous and upcoming seasons.                                            |
| FR-02 | Image-based similarity detection between upcoming and historical products.                                       |
| FR-03 | Ranked recommendation of the most relevant historical items per upcoming product.                                |
| FR-04 | Recommended order quantity per upcoming item, driven by historical order quantity, sell-through, and similarity. |
| FR-05 | User interface presenting comparison results with corresponding product images.                                  |
| FR-06 | Ability to export/download comparison and recommendation results.                                                |

## 7. Expected Benefits

- Data-driven, consistent order quantity decisions in place of purely manual judgement.
- Reduced over-stocking and under-stocking through sell-through-based recommendations.
- Significant time saving in season-on-season product comparison.
- A reusable framework that can be extended to future seasons as data accumulates.

## 8. Assumptions & Dependencies

- Attribute definitions are broadly consistent across seasons; any mapping differences will be resolved during data preparation.
- Image URLs are accessible and correspond correctly to their item codes.
- Historical order quantity and sell-through data are complete and reliable for the last two seasons.
- Business users will validate initial recommendations before adoption.